USB Camera Display

1. Camera Reading

capture=cv.VideoCapture(0)

Parameter meaning:

The parameter in VideoCapture() is 0, which means opening the notebook's built-in camera. If the parameter is a video file path, it opens the video, such as cap = cv2.VideoCapture("../test.avi")

2. Display Camera Video

ret,img = frame.read()

Return value meaning:

ret: ret is a boolean value, used to determine whether a correct frame has been read back

img: image data of each frame

3. Code and Actual Effect Display

Start Docker

After entering the main board desktop, open a terminal and run the following command to start the corresponding Dofbot container:

```
./Docker_Ros.sh
```

Then enter the following command in the Docker terminal to start jupyter-lab,

```
cd
jupyter-lab --allow-root
```

Access Jupyter Lab within Docker:

```
IP:9999         // Example: 192.168.1.11:9999
```

Program path: /root/Dofbot/6.AI_Visual/0.Camera Driver Tutorial.ipynb

Main code:

```
import cv2
import ipywidgets.widgets as widgets
import threading
import time

image_widget = widgets.Image(format='jpeg', width=600, height=500)  #Set camera
display component
display(image_widget)                                               #Display
camera component
```

```
#bgr8 to jpeg format
import enum
import cv2

def bgr8_to_jpeg(value, quality=75):
    return bytes(cv2.imencode('.jpg', value)[1])
```

```
image = cv2.VideoCapture(0,cv2.CAP_V4L2)                    #Open camera
ret, frame = image.read()
image_widget.value = bgr8_to_jpeg(frame)
```

After running the code blocks, the image captured by the USB camera can be displayed.